

# Distributed Training Mechanisms: What Is Sharded and How It Communicates

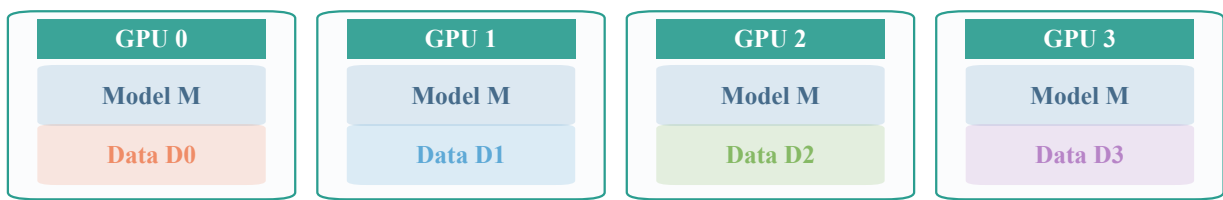
Mechanisms from Table 4-21 plus TP, PP, Ulysses Parallelism and Context Parallelism

## Data Parallel (DDP)

replicate model; split mini-batch

ADOPT

Gradient All-Reduce

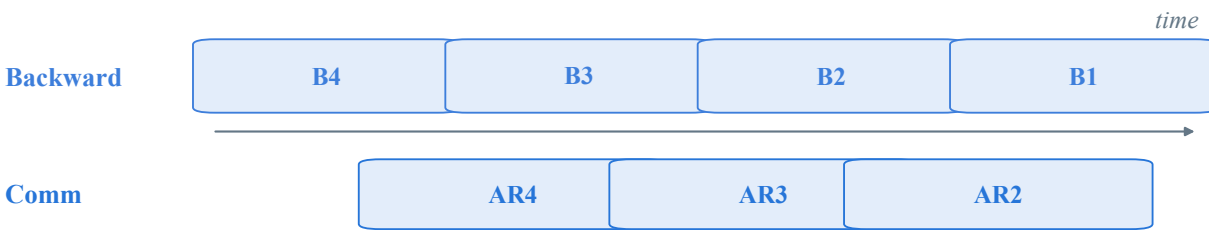


**Shard** batch only; complete model per GPU  
**Comm** bucketed gradient All-Reduce  
**U-MoE** adopt: bucket view + static graph

## Overlap Grad Reduce

launch each bucket when gradients are ready

IN DDP



**Shard** same DDP model and batch layout  
**Comm** async bucket All-Reduce vs backward  
**U-MoE** DDP reducer already provides it

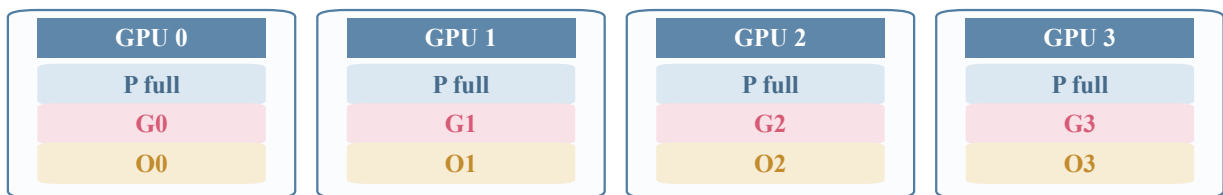
## Distributed Optimizer

Megatron optimizer-state sharding

NOT NOW

Reduce-Scatter G

All-Gather P



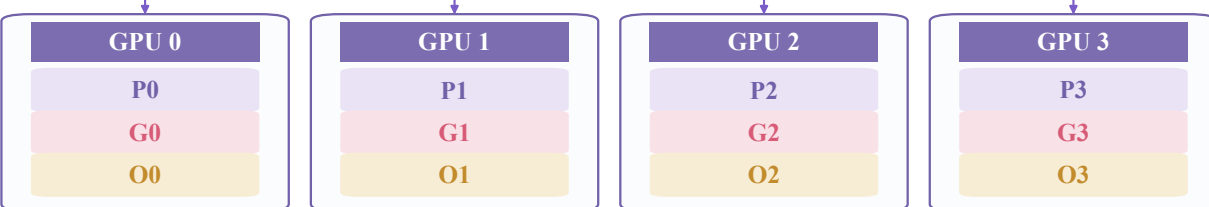
**Shard** optimizer and gradient shards  
**Comm** Reduce-Scatter + parameter All-Gather  
**U-MoE** 4.11 M params: little memory saved

## ZeRO / FSDP

shard P, G and O by stage

NOT NOW

Layer P: All-Gather before compute



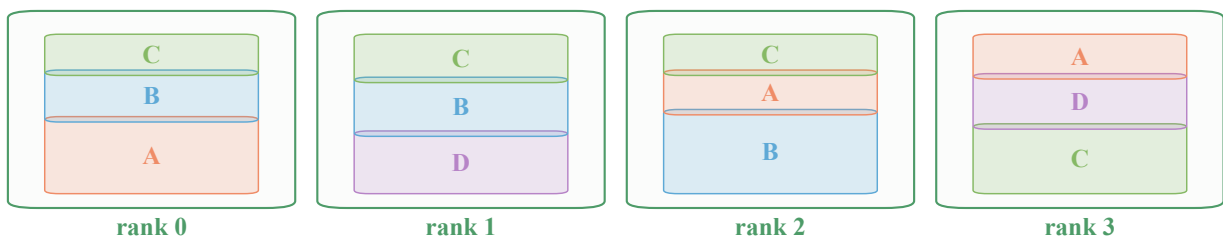
**Shard** parameters / gradients / optimizer  
**Comm** per-layer All-Gather + Reduce-Scatter  
**U-MoE** activation peak dominates; extra collectives

## Cost-Aware Data Shard

equalize predicted work on every rank

PRINCIPLE

equal total predicted cost



**Shard** samples; fixed model on every rank  
**Comm** ordinary DDP after balanced assignment  
**U-MoE** use only with measured cost heterogeneity

## Tensor Parallel (TP)

split matrix dimensions within each layer

NOT NOW

Weight matrix W

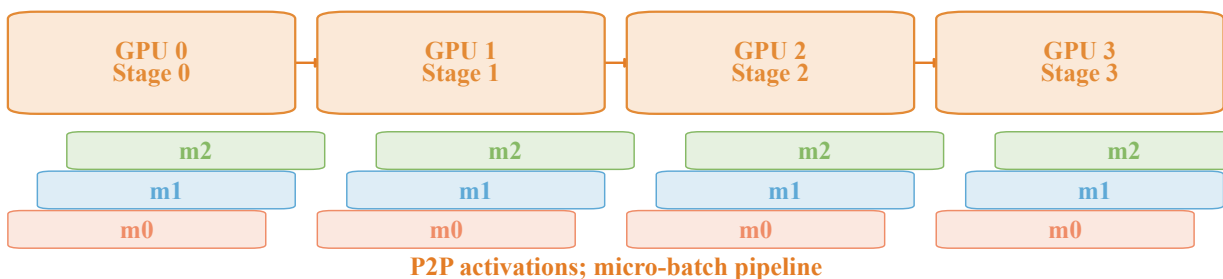


**Shard** rows / columns of every large tensor  
**Comm** All-Reduce or All-Gather each layer  
**U-MoE** C=96 matrices too small for TP

## Pipeline Parallel (PP)

place consecutive layer stages on GPUs

NOT NOW



**Shard** contiguous layer stages  
**Comm** P2P activations and gradients  
**U-MoE** shallow backbone; bubble dominates

## Expert Parallel (EP)

place different experts on different GPUs

NOT NOW

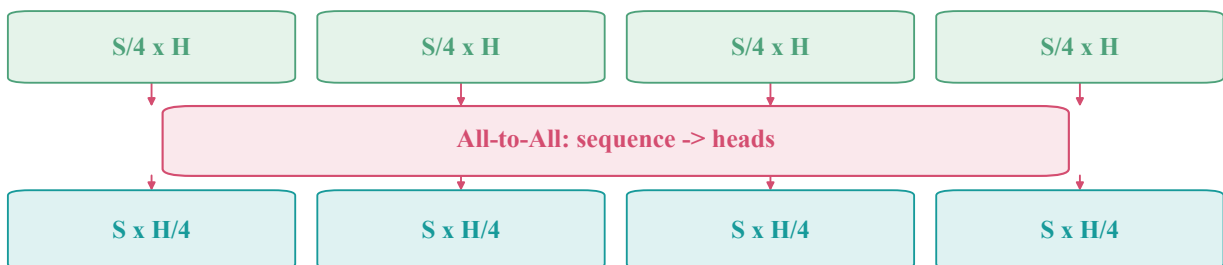


**Shard** expert weights and routed tokens  
**Comm** All-to-All dispatch + combine  
**U-MoE** 12 small experts already fit locally

## Ulysses Parallelism (UP)

swap sequence shards for head shards

NOT NOW

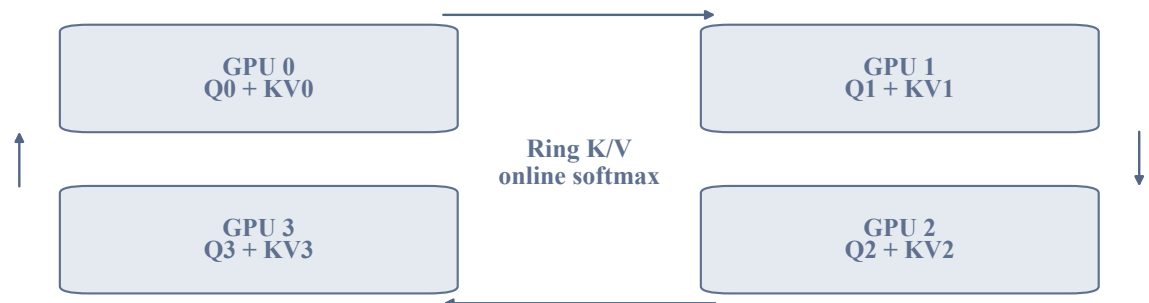


**Shard** sequence first, then attention heads  
**Comm** two All-to-All operations per layer  
**U-MoE** window sequence is only 64 tokens

## Context Parallel (CP)

keep local Q; rotate K/V context blocks

NOT NOW



**Shard** sequence context and K/V blocks  
**Comm** ring P2P K/V exchange  
**U-MoE** local 8 x 8 windows need no CP

## Communication Legend

→ All-Reduce      → Reduce-Scatter / All-Gather      → All-to-All      → P2P      → Ring P2P

## Measured Profile

4.11 M parameters; < 0.1 GB model states  
61-76 GB peak is activation / loss / MoE capacity

## Recommended for U-MoE-Fusion

DDP + bucket view + static graph + 8 MiB buckets + fused Adam  
Cost-aware data assignment only after measured heterogeneity

## Why Other Axes Wait

No parameter, expert-capacity, depth, or long-context bottleneck  
Extra collectives would target the wrong resource